

Practical No. 8

VPC settings

Resources to create

Info

Create only the VPC resource or the VPC and other networking resources.

☐ VPC only

☒ VPC and more

Name tag auto-generation

Info

Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.

☒ Auto-generate

lab

IPv4 CIDR block

Info

Determine the starting IP and the size of your VPC using CIDR notation.

10.0.0.0/16

65,536 IPs

1.2

Number of Availability Zones (AZs)

Info

Choose the number of AZs in which to provision subnets. We recommend at least two AZs for high availability.

1

2

3

Customize AZs

Number of public subnets

Info

The number of public subnets to add to your VPC. Use public subnets for web applications that need to be publicly accessible over the internet.

0

1

2

Number of private subnets

Info

The number of private subnets to add to your VPC. Use private subnets to secure backend resources that don't need public access.

0

1

2

Customize subnets CIDR blocks

Public subnet CIDR block in us-east-1a

10.0.0.0/24

256 IPs

Private subnet CIDR block in us-east-1a

10.0.1.0/24

256 IPs

1.2

NAT gateways (\$)

Info

Choose the number of Availability Zones (AZs) in which to create NAT gateways. Note that there is a charge for each NAT gateway

In 1 AZ

1 per AZ

VPC endpoints

Info

Endpoints can help reduce NAT gateway charges and improve security by accessing S3 directly from the VPC. By default, full access policy is used. You can customize this policy at any time.

None

S3 Gateway

1.2

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

lab-subnet-public2

The name can be up to 256 characters long.

Availability Zone

Info

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (N. Virginia) / use1-az4 (us-east-1b)

IPv4 VPC CIDR block

Info

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.2.0/24

2.1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

lab-subnet-private2

The name can be up to 256 characters long.

Availability Zone

Info

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (N. Virginia) / use1-az4 (us-east-1b)

IPv4 VPC CIDR block

Info

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.3.0/24

2.2

Available subnets (2/4)						
Filter subnet associations						
	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID	
☐	lab-subnet-public2	subnet-06d369d33722fb44e	10.0.2.0/24	-	Main (rtb-0ed0480e3d2783089)	
☒	lab-subnet-private1-us-east-1a	subnet-0080d7bd31881b4da	10.0.1.0/24	-	rtb-03989d4f8310791e9 / lab-rtb-private1-us-east-1a	
☐	lab-subnet-public1-us-east-1a	subnet-00676d824d0ceb8d8	10.0.0.0/24	-	rtb-0f735ebff3dbdbfb4 / lab-rtb-public	
☒	lab-subnet-private2	subnet-084d6fa7b12873aa2	10.0.3.0/24	-	Main (rtb-0ed0480e3d2783089)	

2.3

Available subnets (2/4)						
Filter subnet associations						
	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID	
☒	lab-subnet-public2	subnet-06d369d33722fb44e	10.0.2.0/24	-	Main (rtb-0ed0480e3d2783089)	
☐	lab-subnet-private1-us-east-1a	subnet-0080d7bd31881b4da	10.0.1.0/24	-	rtb-03989d4f8310791e9 / lab-rtb-private1-us-east-1a	
☒	lab-subnet-public1-us-east-1a	subnet-00676d824d0ceb8d8	10.0.0.0/24	-	rtb-0f735ebff3dbdbfb4 / lab-rtb-public	
☐	lab-subnet-private2	subnet-084d6fa7b12873aa2	10.0.3.0/24	-	rtb-03989d4f8310791e9 / lab-rtb-private1-us-east-1a	

2.4

Basic details

Security group name

Info

Web Security Group

Name cannot be edited after creation.

Description

Info

Enable HTTP access

VPC

Info

vpc-0a74f17a090aac752 (lab-vpc)

3.2

Inbound rules

Info

Type

Info

HTTP

Protocol

Info

TCP

Port range

Info

80

Source

Info

Anywh...

Description - optional

Info

Permit web requests

3.3

Name and tagsInfo

Name

Web Server 1

Application and OS Images

An AMI contains the operating system, applications, and data. You can select an AMI from the catalog or choose **Browse more AMIs**.

Search our full catalog including 1000s of AMIs

AMI from catalog

Recents

Name

Amazon Linux 2023 kernel-6.1 AMI

4.1

Instance typeInfo | Get advice

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current On-Demand Linux base pricing: 0.0116 USD per hour On-Demand Windows base pricing: 0.0162 USD per hour On-Demand Ubuntu Pro base pricing: 0.0134 USD per hour On-Demand SUSE base pricing: 0.0116 USD per hour

Additional costs apply for AMIs with pre-installed software

Key pair (login)Info

You can use a key pair to securely connect to your instance.

Key pair name - required

vockey

4.1

Network settingsInfo

VPC - requiredInfo

vpc-0a74f17a090aac752 (lab-vpc)

10.0.0.0/16

SubnetInfo

subnet-06d369d33722fb44e

lab-subnet-public2

VPC: vpc-0a74f17a090aac752 Owner: 381492002005 Availability Zone: us-east-1b (use1-az4) Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.2.0/24

Auto-assign public IPInfo

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups)Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic.

Create security group

Select existing security group

Common security groupsInfo

Select security groups

Web Security Group sg-06b19352f580564fb X

VPC: vpc-0a74f17a090aac752

4.2

User data - optionalInfo

Upload a file with your user data or enter it in the field.

Choose file

#!/bin/bash

Install Apache Web Server and PHP

dnf install -y httpd wget php mariadb105-server

Download Lab files

wget https://aws-tc-largeobjects.s3.us-west-2.amazonaws.com/CUR-TF-100-ACCLFO-2/2-lab2-vpc/s3/lab-app.zip

unzip lab-app.zip -d /var/www/html/

Turn on web server

chkconfig httpd on

service httpd start

4.3

Not secure 54.87.230.152

aws

Load Test

RDS

Meta-Data

Value

InstanceId

i-0696645b3e177bcd

Availability Zone

us-east-1b

Current CPU Load: 1%

4.4